

# Housing Returns over the Income Distribution

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# Housing is the largest asset, and it is not a stock

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- Return heterogeneity is a core driver of wealth inequality  
Fagereng et al. (2020), Bach, Calvet and Sodini (2020)
- Housing is 53% of gross wealth in Denmark and dominates portfolios below the top
- Existing work measures housing returns with **aggregate price indices** plus a **uniform rental yield**
  - All variation in housing returns then comes from price growth
- But housing is not a stock
  - Lumpy, indivisible, tied to one location
  - Bought with leverage under LTV and PTI limits
  - Bundles an investment with a consumption good

**Explanations based on risk-taking, skill or scale need not carry over to housing.**

# This paper

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**Question:** Does the return to housing differ across the income distribution, and why?

**Data:** Danish administrative registers, 1996–2022, 218,000 repeat sales linked to buyer income

## Findings

1. Higher-income buyers earn higher capital gains: **1.36 pp per year** from P10 to P90  
Location statistically accounts for nearly all of the gap
2. Rental yields fall just as steeply, so **total returns are roughly equal**  
The *composition* differs: gains at the top, housing dividend at the bottom
3. Lower-income buyers face much smaller feasible sets in high-growth areas  
Credit reforms moved prices, not the income composition of buyers

# Data and measurement

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## Registers (Statistics Denmark)

- All transactions with exact dates, prices, attributes
- Linked income, wealth, demographics via CPR
- Income rank within year and age cohort, 3-year average
- Primary residences, buyers aged 25+

## Rental yields

- Postcode rent/price ratio from 1999 BBR rents
- Validated against national accounts and 2015–22 private rental data

## Unlevered real capital gain

$$r_i^u = \frac{\ln P_{is} - \ln P_{ip}}{T_{is} - T_{ip}}$$

## Total return

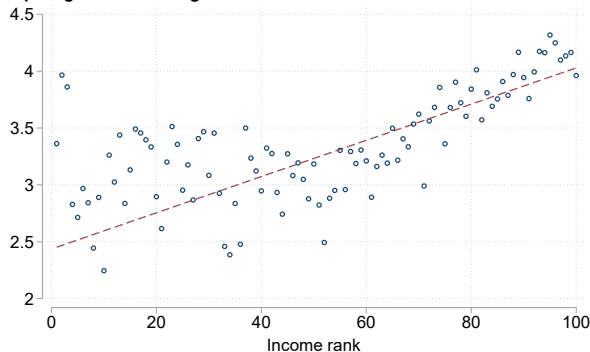
- Additive:  $r_i^u$  + location yield
- IRR: nets out property tax, maintenance (1.5%) and transaction costs

## Specification

$$Y_i = \beta_0 + \beta_1 \text{IncRank}_i + X_i \Gamma + \epsilon_i$$

# Capital gains rise steeply with buyer income rank

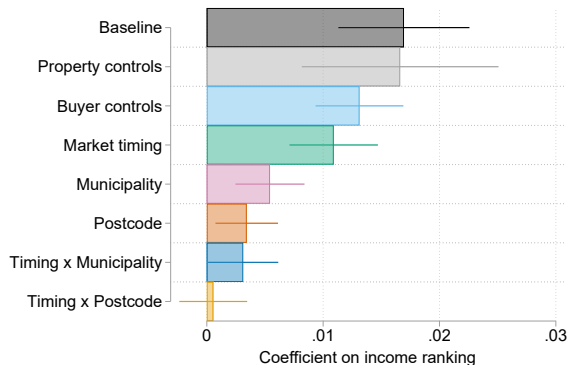
Capital gains to housing



- Baseline slope  $\hat{\beta}_1 = 0.017$  (s.e. clustered by municipality)
- P90 – P10 gap: 1.36 pp per year
- Over a 10-year hold:  $\approx 14.6\%$  higher cumulative gain
- Holding periods are similar across groups (8 years)

Annualized log real capital gains, repeat sales. Each dot is a rank-bin mean.

# Location accounts for the capital-gains gap



- Property and buyer controls: 0.017  $\rightarrow$  0.013
- National market timing raises  $R^2$  from 0.04 to 0.23 but barely moves  $\hat{\beta}_1$
- Municipality FE: 0.005  
Postcode FE: 0.003
- Timing  $\times$  postcode:  $\approx 0$
- Gelbach decomposition: location explains 67–88% of the change

**A control matters only if it predicts gains *and* covaries with income. Location does both.**

## Within a market, the income gradient turns negative

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| Market                       | $\hat{\beta}_1$ |
|------------------------------|-----------------|
| Capital area                 | -0.004          |
| Capital + 3 major cities     | -0.003          |
| 8 largest cities             | -0.003          |
| 7 largest outside Copenhagen | -0.004          |

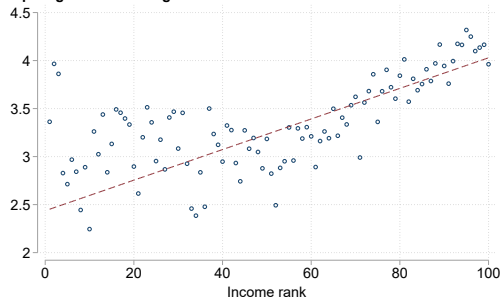
All significant at 5% or better. Postcode, purchase-year, sale-year FE and controls.

- Higher-income buyers do **not** pick better properties within a market
- Evidence against skill, information or sophistication as the channel
- Risk differences across locations are also small

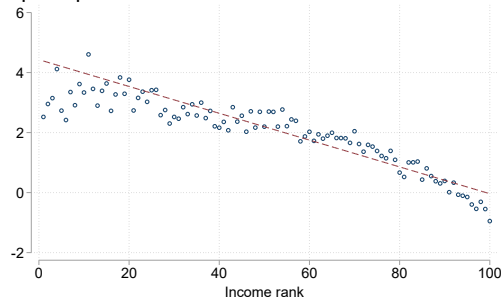
**The national gradient reflects exposure to appreciating markets. So why do buyers sort?**

# Rental yields fall as steeply as capital gains rise

Capital gains to housing



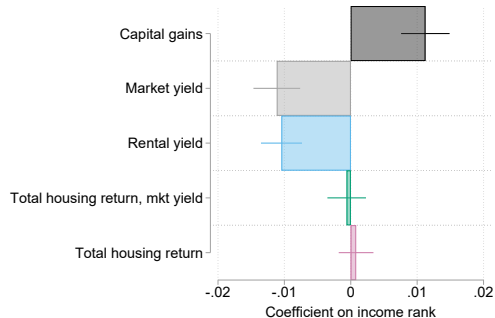
Implicit imputed rents



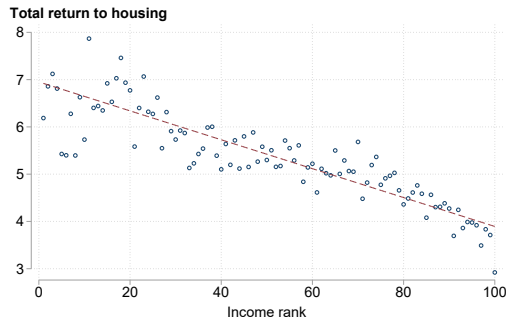
- Expensive, fast-appreciating markets have low rents relative to prices
- Higher-income buyers concentrate there, so the yield gradient mirrors the gain gradient

Right panel: implied net rental yield, IRR minus capital gain. No controls.

# Total housing returns are roughly equal



(a) Additive: gain + location yield



(b) Internal rate of return

- Additive measure: total-return gradient  $\approx 0$ , stable across specifications
- IRR, net of carrying costs: gradient of  $-0.031$ , modestly *negative*

## Equal returns, unequal wealth accumulation

Total return:  $r_{it}^H = g_{it} + h_{it}$

$$\Delta W_{it} = g_{it}H_{it} + G_{it}^O + S_{it}^A$$

- Capital gain  $gH$  enters the balance sheet directly, largely untaxed
- Rental yield  $hH$  is a flow of housing services, consumed
- The yield does not appear in  $\Delta W$  unless households save more ( $\lambda > 0$ )

P90 – P10 gap in wealth returns (pp)

|           | Gains | Add.  | IRR   |
|-----------|-------|-------|-------|
| Housing   | 1.24  | 1.32  | 0.32  |
| Financial | 0.41  | 0.41  | 0.41  |
| Pension   | −0.60 | −0.60 | −0.60 |
| Total gap | 1.06  | 1.14  | 0.13  |

Decomposition following Kuhn, Schularick and Steins (2020). Calibration, order of magnitude.

**Housing is the largest single contributor to the wealth-return gap under two of three measures.**

# Why do income groups buy in different places?

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## A feasible choice set, bounded on two sides

- From above: financial constraints

$$P^{\max} = M^{\max} + W, \quad M^{\max} = \min \left( \bar{M}^{LTV}, \bar{M}^{PTI} \right)$$

- LTV: 20% down payment from liquid assets and home equity
- PTI: mortgage payment  $\leq$  35% of income, 30-year annuity

- From below: consumption needs

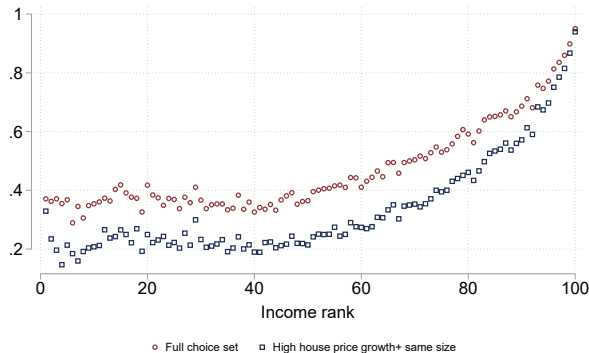
- A household cannot buy a fraction of a high-growth neighborhood

## Two predictions

1. Feasible sets shrink with income, more so in high-growth areas at a given size
2. If constraints are the barrier, loosening credit should move lower-income buyers into high-growth areas

# Low-income buyers can afford little where gains are high

Feasible transaction share



- Share of same-year transactions priced at or below  $P^{\max}$
- Bottom third:  $\approx 40\%$   
Top third:  $> 60\%$
- High-growth areas, same dwelling size: about  $20\%$  at P40
- PTI binds for over half of buyers and drives the gap

Red: full choice set. Blue: top-quintile growth municipalities, size within 10%. 25% sample.

## Test: 2003 interest-only loans loosened PTI limits

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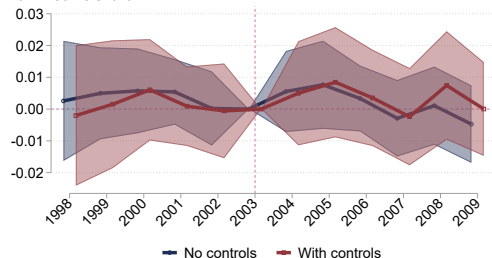
- Interest-only loans cut monthly payments by roughly 20%
- Uptake above 60% of purchases across the income distribution

$$Y_{imt} = \alpha_m + \gamma_t + \sum_{k \neq 2003} \beta_k (\text{HighGrowth}_m \times \mathbf{1}_{t=k}) + X_{it}\Gamma + \epsilon_{imt}$$

- $\text{HighGrowth}_m$ : top quintile of municipal price growth, 1997–2002
- $Y$ : indicator for a lower-third income buyer, or log price
- Municipality, year and region  $\times$  year FE, clustered by municipality
- Identification: parallel pre-trends in *buyer composition*

# Looser credit raised prices, not low-income entry

Low income share



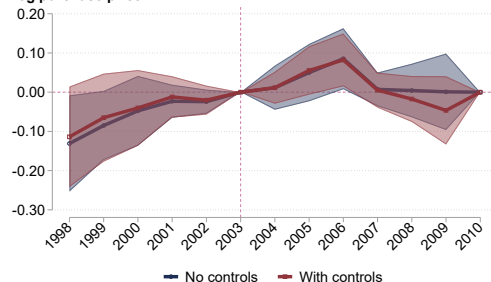
Share of low income buyers 1998-2002

All transactions: 7.841%. High-return municipalities.: 6.499%

Pre-trend F-test:  $F(\text{No controls})=0.602$  ( $p=0.699$ ),  $F(\text{Controls})=0.526$  ( $p=0.756$ )

(a) Share of lower-income buyers

Log purchase price



Pre-trend F-test:  $F(\text{No controls})=5.101$  ( $p=0.000$ ),  $F(\text{Controls})=6.951$  ( $p=0.000$ )

(b) Log purchase price

- Pooled post-reform effect 0.003 (s.e. 0.003). Rules out a rise above 0.9 pp, 12% of the 7.8 pp mean
- The 2016 tightening in Copenhagen and Aarhus also left composition unchanged

## Re-sorting happens only where supply is elastic

| Buyer income group          | Low               | Middle               | High                |
|-----------------------------|-------------------|----------------------|---------------------|
| Post $\times$ HighGrowth    | -0.009<br>(0.021) | 0.127**<br>(0.048)   | -0.174**<br>(0.078) |
| $\times$ Inv. supply elast. | 0.009<br>(0.015)  | -0.102***<br>(0.036) | 0.142**<br>(0.058)  |

Supply elasticity following Guren et al. (2021).  $N = 393,352$ .

- Middle-income share rose in high-growth areas *with elastic supply*
- The effect shrinks as supply becomes more constrained
- Consistent with Greenwald and Guren: with tight supply, credit is capitalized into prices

**Constraints set baseline access. Inelastic supply keeps looser credit from changing who buys.**

# Conclusion

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- Higher-income buyers earn higher housing capital gains, and location accounts for nearly all of the gap
- Lower rental yields in those locations offset the gains. Total returns are roughly equal
- What differs is the **composition**: wealth-building gains at the top, consumed housing services at the bottom
- Borrowing limits and inelastic supply act as a **participation constraint** in high-growth markets

## Caveats

- Realized, not expected, returns over 1996–2022
- Location fixed effects capture both preferences and constraints. We test channels but do not isolate one

**Thank you**

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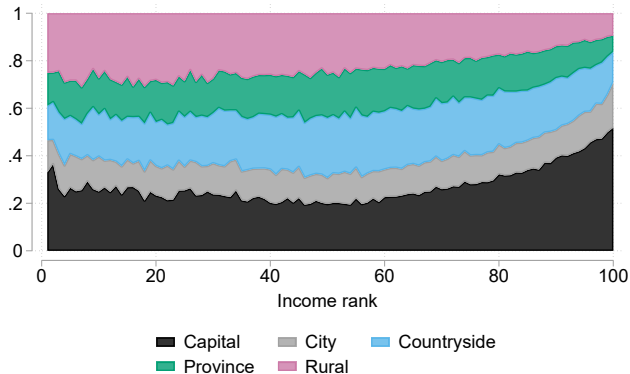
**Backup slides**

## Summary statistics by income tercile

|                                  | All       | Low     | Middle    | High      |
|----------------------------------|-----------|---------|-----------|-----------|
| Purchase price (DKK)             | 1,205,815 | 953,824 | 1,036,606 | 1,360,897 |
| Years between transactions       | 8         | 8       | 8         | 8         |
| Annualized real capital gain (%) | 3.6       | 3.2     | 3.1       | 4.0       |
| Capital                          | 5.2       | 5.6     | 5.3       | 5.1       |
| Province                         | 2.8       | 2.0     | 2.3       | 3.3       |
| Rural                            | 2.4       | 2.0     | 2.1       | 3.0       |
| Apartment                        | 0.31      | 0.35    | 0.24      | 0.34      |
| Size (m <sup>2</sup> )           | 112       | 101     | 112       | 113       |
| Income, pre-purchase (DKK)       | 324,452   | 125,490 | 220,140   | 427,337   |
| Wealth rank                      | 46        | 50      | 39        | 50        |
| Age                              | 41        | 45      | 38        | 42        |
| Renovation indicator             | 0.34      | 0.26    | 0.29      | 0.39      |
| Share of all transactions        |           | 0.09    | 0.36      | 0.55      |
| Repeat-sale share                | 0.27      | 0.31    | 0.27      | 0.26      |
| <i>N</i>                         | 217,926   | 19,826  | 79,329    | 118,771   |

# Higher-income buyers concentrate in large cities

Share of buyers by urbanization level



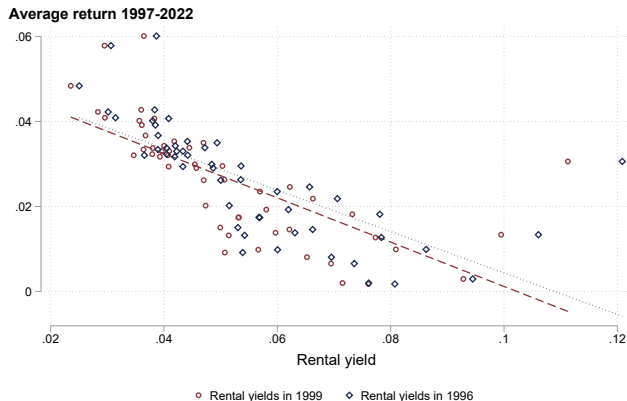
## Internal rate of return

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$$-P_{ip}(1 + \kappa) + \sum_{t=1}^{H_i-1} \frac{\text{NetRent}_{it}}{(1 + r_i^{\text{total}})^t} + \frac{\text{NetRent}_{iH_i} + P_{is}(1 - \tau)}{(1 + r_i^{\text{total}})^{H_i}} = 0$$

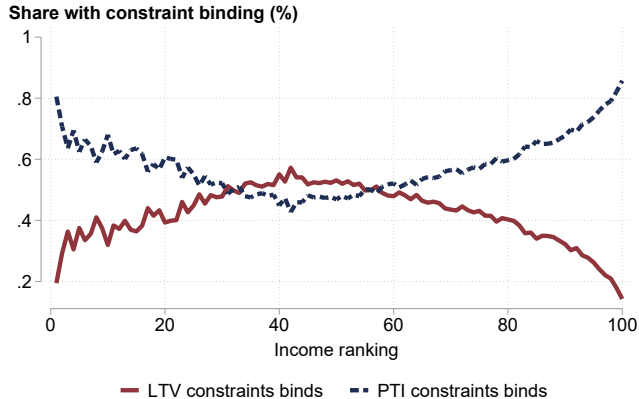
- Gross rent: postcode yield  $\times$  property value, value interpolated linearly
- Net rent subtracts observed property tax and 1.5% maintenance
- Closing costs  $\kappa = 0.6\%$ , sale costs  $\tau = 4\%$ , following Kermani and Wong (2021)
- Implied net yield  $= r_i^{\text{total}} - r_i^u$

# Rental yields are lower where prices grow faster



Postcode rental yield from 1999 BBR against average capital gains.

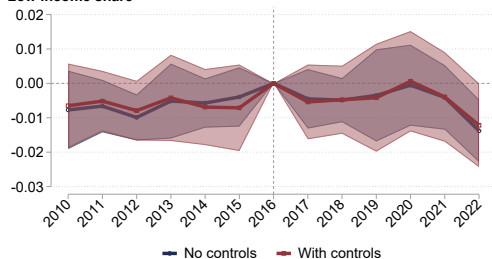
# PTI is the binding constraint for most buyers



LTV with 20% down payment. PTI with 35% of income and a 30-year annuity.

# 2016 guidance: tighter credit, same composition

Low income share



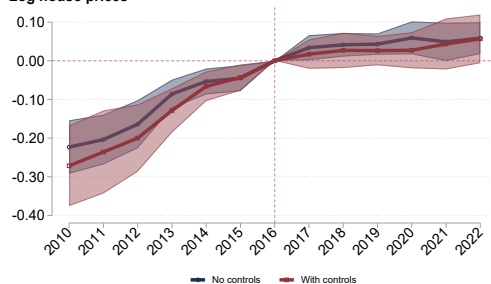
Share of low income buyers 2010-2015

All transactions: 6.626%. Treated municipalities.: 5.348%

Pre-trend F-test:  $F(\text{No controls})=1.707$  ( $p=0.127$ ),  $F(\text{Controls})=1.368$  ( $p=0.235$ )

(a) Share of lower-income buyers

Log house prices



Pre-trend F-test:  $F(\text{No controls})=9.603$  ( $p=0.000$ ),  $F(\text{Controls})=14.737$  ( $p=0.000$ )

(b) Log house prices

FSA guidance for Copenhagen, surrounding municipalities and Aarhus: stressed rates and stricter assessments for economically sensitive borrowers.

# Housing drives the P90–P10 wealth-return gap



Capital-gains-only measure. Composition, within and interaction terms.